

Bio-Pulse Symbiosis

智脉共生



Bio-Physarum Mobility Network Renewal for the Dazhongsi AI Industry Cluster

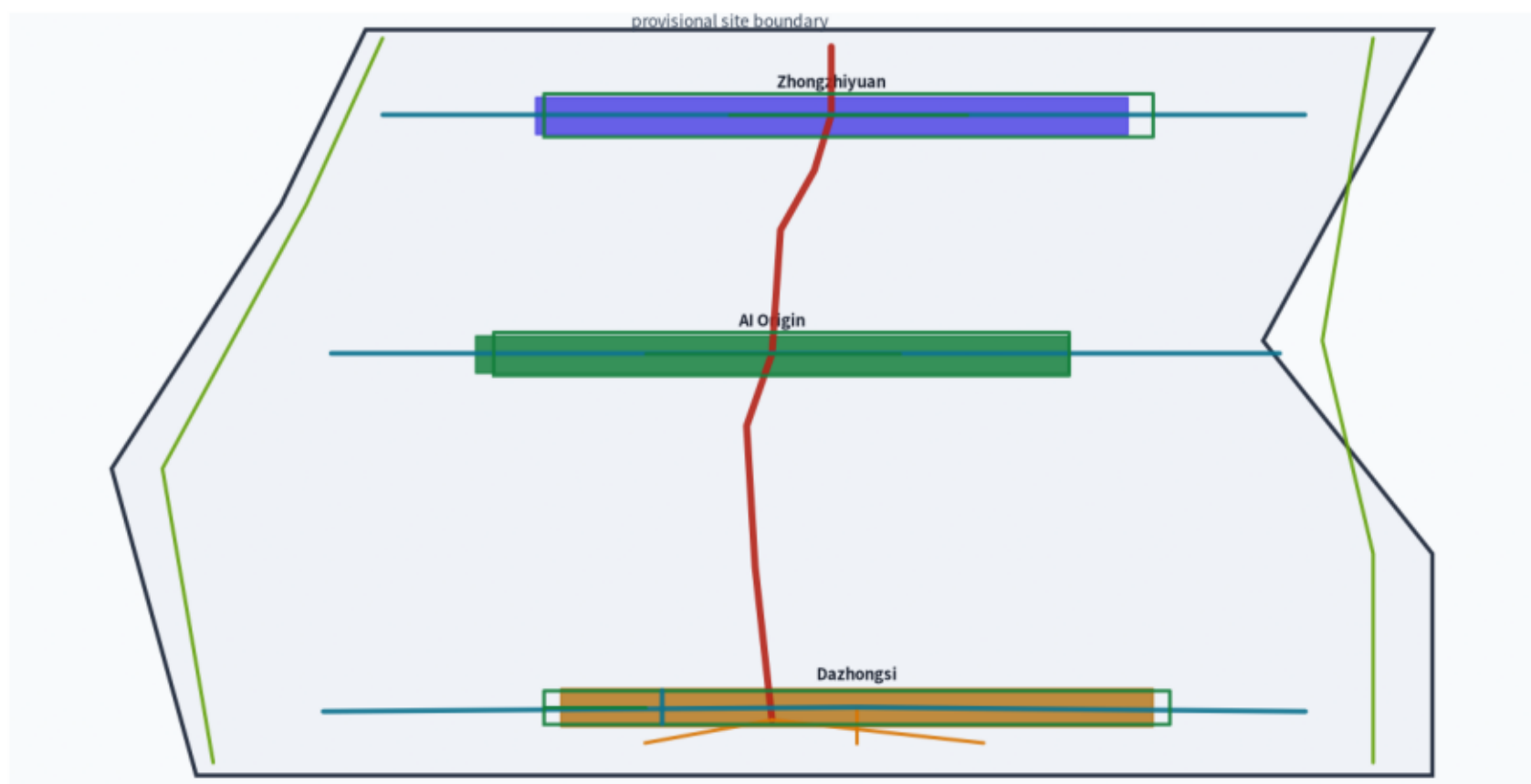
Method: Bio-Physarum (Tero et al. 2010) + NSGA-II

Validation: 167-edge / efficiency 19.20 / heritage $f_3 \equiv f_2$

Recommended: Plan03 UDS 80.34 • permeable 69.1%

Formal geometry = conceptual network inside provisional boundary; real Physarum is method-validation evidence only.

Evidence chain and package



Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective | Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective $f_3 \equiv f_2$ (degenerates to cost; no independent site heritage conclusion)

Three-level scope and spatial structure

Coordinated study scope

43.6 km² AI ecosystem / three zones two wings / future urban form

Overall design scope

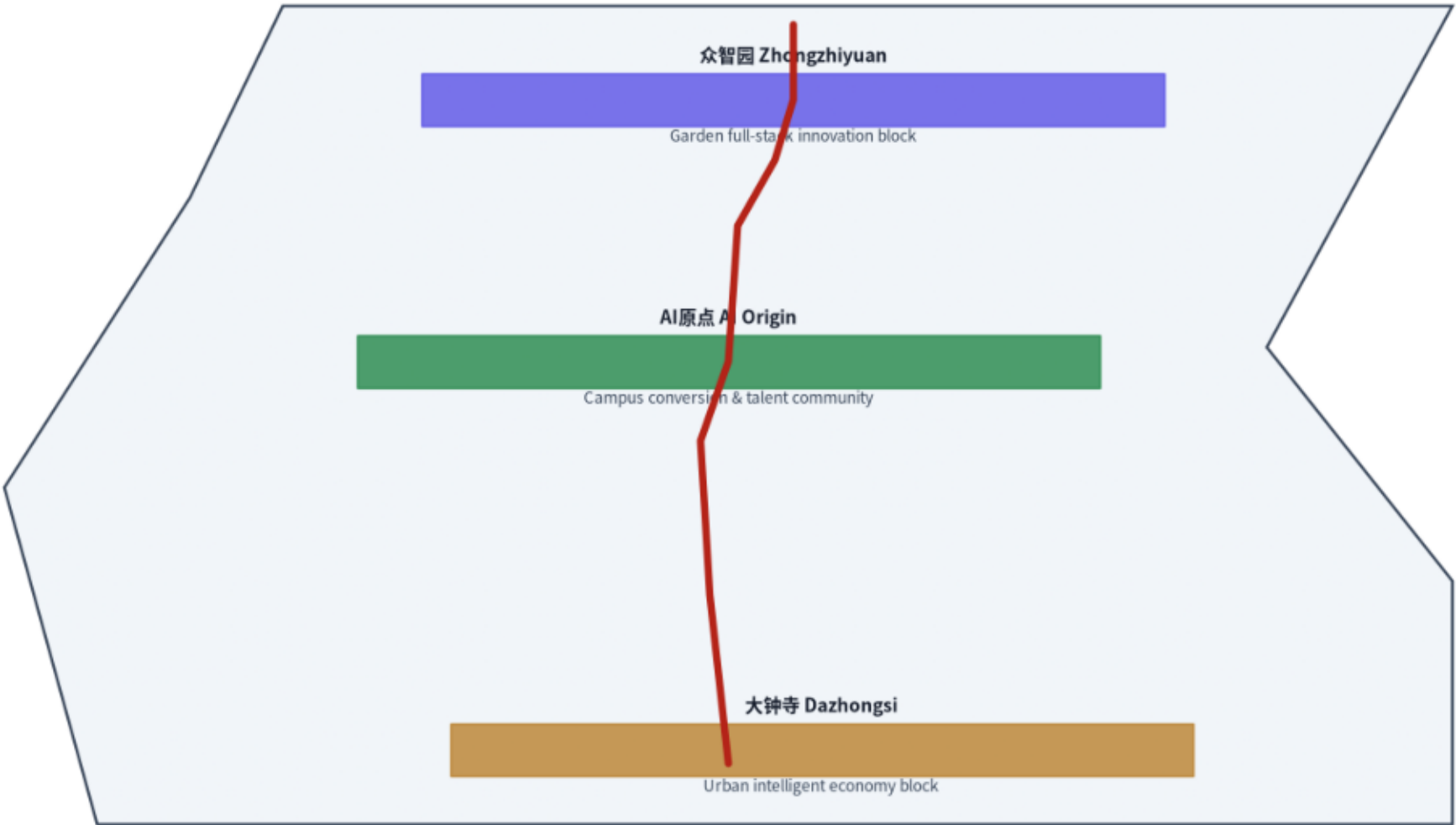
11.4 km² Urban renewal framework / land use / transport & utilities

Key area scope

368.4 ha Three key detailed-design areas

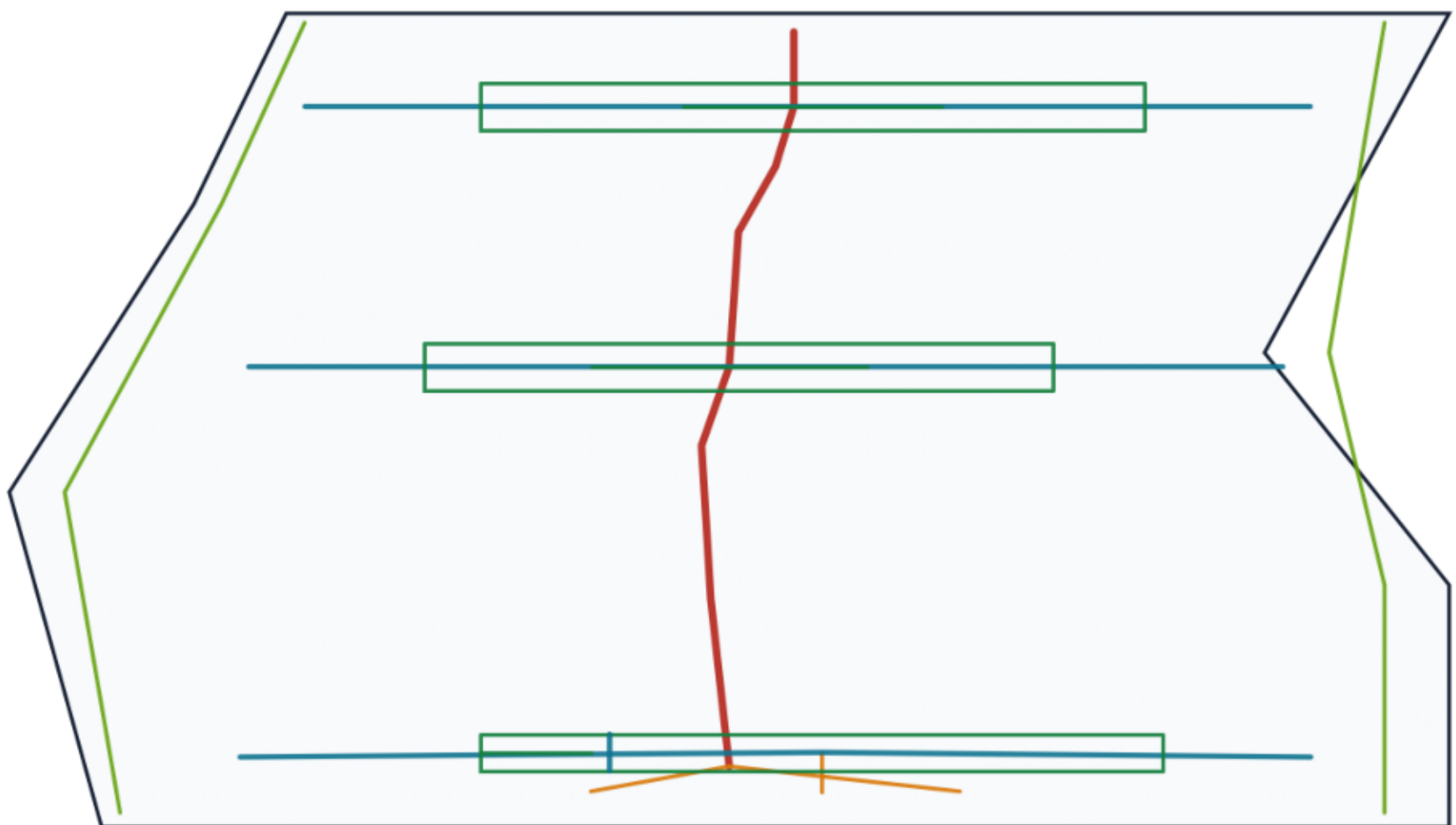
Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

Key areas and Physarum spine



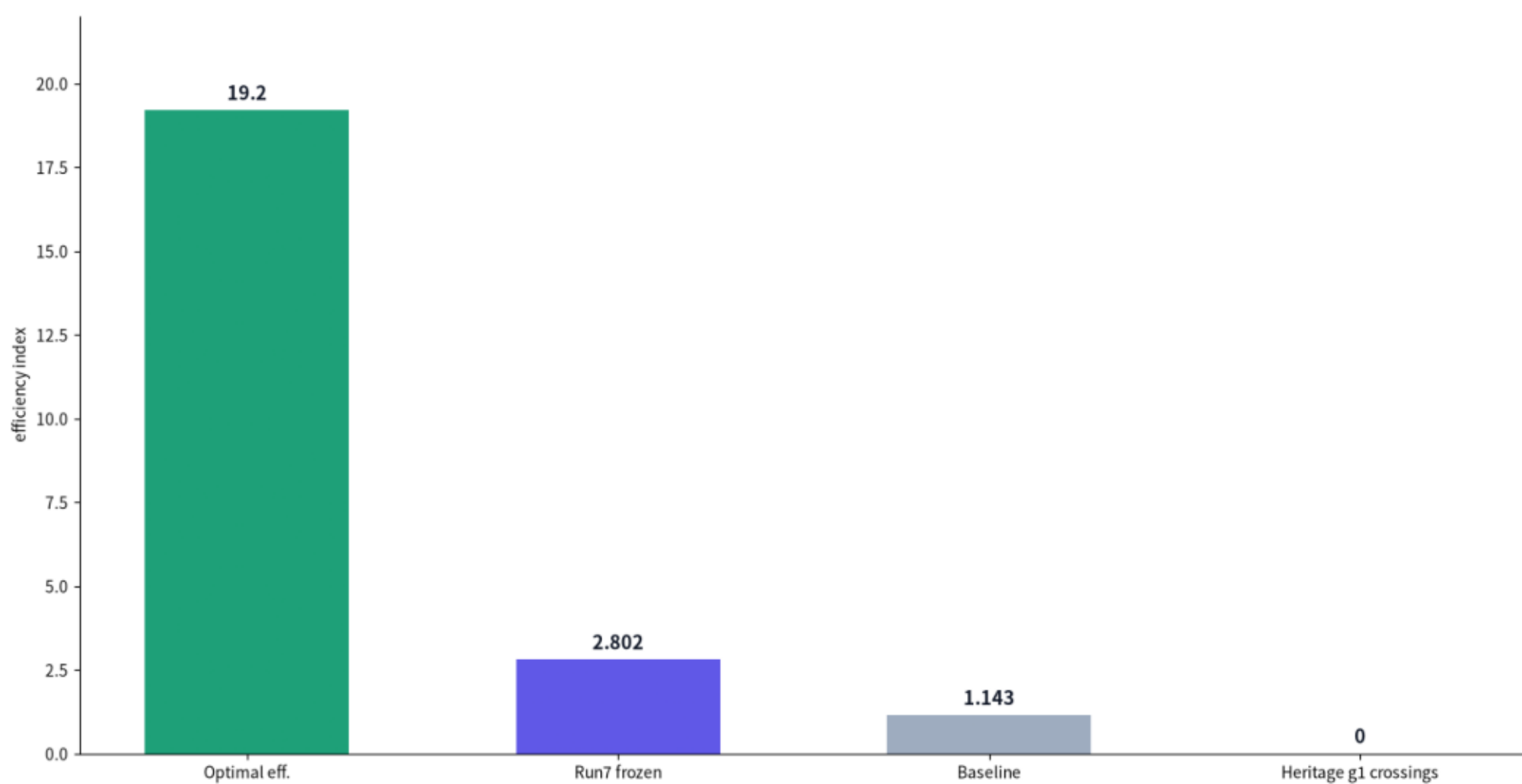
Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

Mobility and blue-green public space



Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

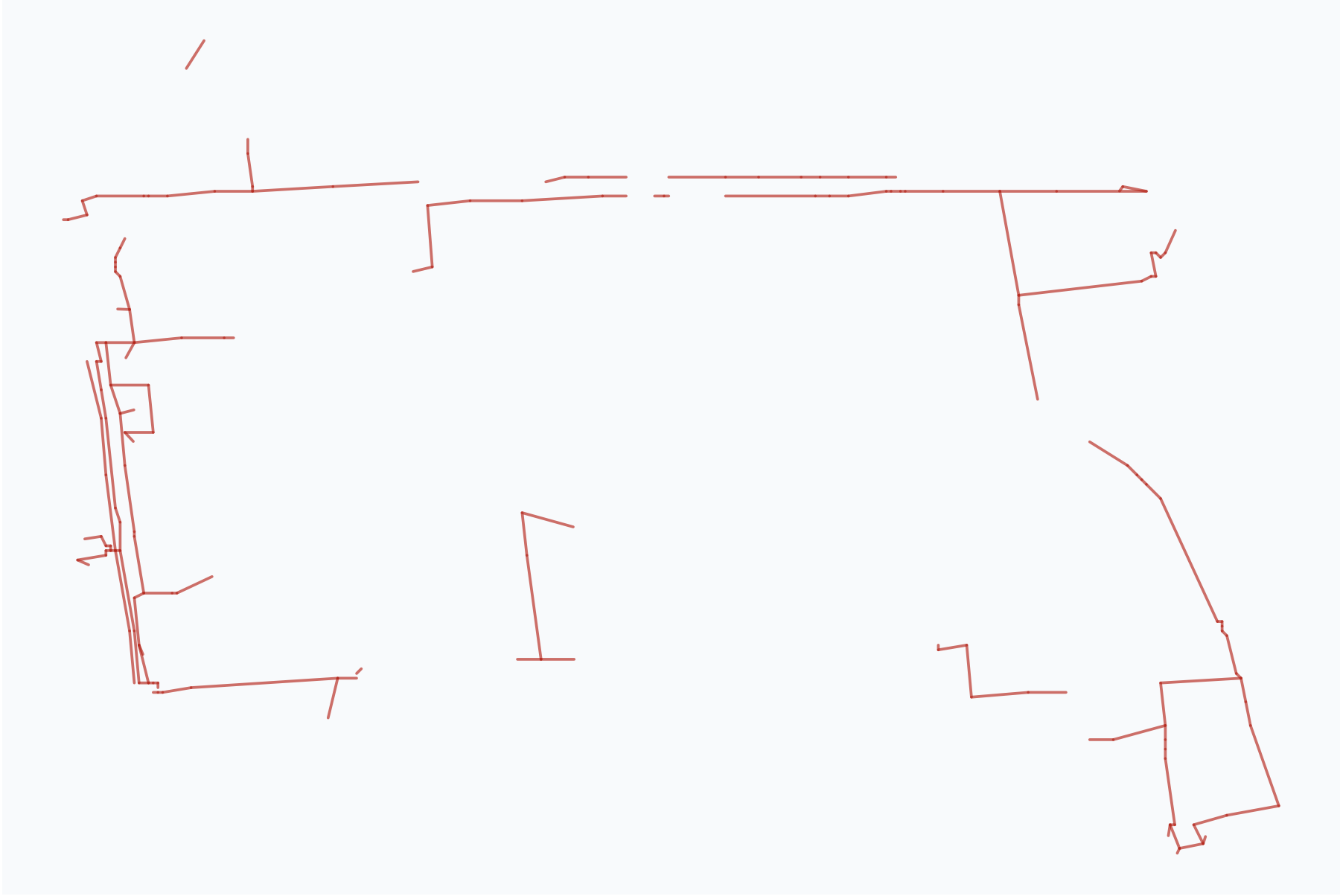
Metrics and method validation



Plan03 urban-integration UDS 80.34 recommended; heritage objective $f_3 \equiv f_2$ degenerates to cost — '0 heritage crossings' is g1 hard-constraint behavior only, not an independent site heritage conclusion
Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective $f_3 \equiv f_2$ (degenerates to cost; no independent site heritage conclusion)

Brand: 智脉共生 (Bio-Pulse Symbiosis)

Real Physarum 167-edge skeleton (method validation)



Zhi = AI belt Mai = Physarum network Symbiosis = blue-green • slow • industry • community

Conceptual brand (suggestive framework); does not replace existing names/trademarks/marks.

Visual base: real Physarum run skeleton (167 edges, ~2-3 km west of provisional boundary, method validation only).

Global Benchmark Cases (Public Information)

案例 Case	地点 Location	核心做法 (公开报道)	借鉴 / 警示
Sidewalk Labs Quayside	Toronto, CA	Sensing, adaptive building, data trust (2017-2020)	Governance mechanism / privacy caution
Masdar City	Abu Dhabi, UAE	Low-emission block, passive cooling, early PRT	Low-carbon + feeder / fiscal balance
Songdo IBD	Incheon, KR	Reclaimed town, U-City sensing, pneumatic waste	New infra reservation / keep organic growth
Amsterdam Smart City	Netherlands	PPP collaboration, living-lab pilots	Incremental + multi-party / responsibility
Xiong'an New Area	Hebei, CN	CIM twin, blue-green network, utility tunnels	Digital twin + blue-green slow priority

Test Validation Scenarios (suggestive framework)

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场景	验证对象	检验依据（真实）	关键目标
T-01 Dazhongsi TOD	Four-quadrant link & feeder	GB 50763-2012 / GB/T 51328-2018	Barrier-free flow; feeder on primary vein
T-02 Xiaoyue stormwater	Blue-green + permeable paving	Sponge City guide / GB 50014	Permeable 69.1% / green pen. 25.2%
T-03 AI Origin access	Campus-park-block stitching	GB 50763-2012	Slow loop covers nodes; continuous access

Implementation Matrix (suggestive framework)

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项目	建议责任主体	资金来源（建议性）	建议周期	暂停/退出条件
JZ-01 Slow-travel stitching	Transport commission + park	Fiscal + renewal fund	Near-term	Pause if ownership unsettled
JZ-02 Qinghe interface	Water authority + park	Fiscal + green bond	Mid-term	Exit if flood unmet
JZ-03 Conversion street	University + sci-tech bureau	University + social capital	Mid-term	Pause if ownership dispute
JZ-04 Dazhongsi four-quadrant	Rail operator + transport	Rail + fiscal	Mid-term	Postpone if timing mismatch
JZ-05 Edge computing	Sci-tech bureau + operator	Social capital + subsidy	Long-term	Exit if operator absent
JZ-06 Physarum recalibration	Design team + university	Research funding	Long-term	Keep 'to be confirmed'

Responsible bodies, funding, cycles are suggestive; not official approval or funding commitment.

Accessibility & Inclusivity (GB 50763-2012)

画像

Open-source developer

Startup team

Leading-enterprise visitor

Nearby resident

University teacher/student

无障碍与包容性需求

Night collaboration reachability & safety

Members with disabilities

International / reduced-mobility

Elderly, children, disabled daily passage

Disabled campus-park connectivity

空间响应

Barrier-free entrance/toilet, graded lighting

Barrier-free office/test space, lowered desk

Multilingual signage, barrier-free elevator

Curb ramps, tactile paving, rest seats

Barrier-free slow stitching, experience point